

Mahesh Chitakoti

+91 94817 77227 | maheshchitkoti@gmail.com | LinkedIn | GitHub

Summary

Backend & AI Engineer with **2.5+ years of experience** architecting high-throughput microservices, RAG systems, and event-driven GenAI pipelines. Specialized in **Python, FastAPI, TypeScript, Qdrant, and distributed caching**, with a proven track record of optimizing LLM token economics, reducing database latency, and delivering resilient, production-grade cloud APIs.

Technical Skills

Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, NestJS, Express.js, AsyncIO, REST APIs, Microservices, WebSockets, MQTT

GenAI & Orchestration: RAG, Agentic Workflows (LangChain, LangGraph), Tool Calling, Structured Outputs (Pydantic), Semantic Caching, Prompt Evaluation

Vector Systems & Retrieval: Qdrant, pgvector, Hybrid Search (Dense + BM25), Cross-Encoder Reranking, Embeddings Pipeline Optimization

Cloud, DevOps & Data: PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD, Google Gemini APIs, OpenAI API

Experience

AI Engineer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

Remote

- Architected an asynchronous GenAI pipeline using **FastAPI, Google Gemini APIs, and Qdrant**; implemented **semantic caching, Pydantic schema validation, and token-bucket rate limiting**, cutting redundant inference costs by **35%** and eliminating rate-limit (429) failures.
- Developed semantic-search microservices over lesson transcripts in **Qdrant**, indexing **10,000+ chunks** with payload metadata filtering to achieve **300ms p95 retrieval latency** for automated CEFR language proficiency grading.
- Engineered full-duplex **WebSocket and MQTT** ingestion pipelines with **exponential backoff, dead-letter queues (DLQ), and automatic session recovery**, maintaining resilient real-time audio streaming across high-latency client connections.
- Instrumented backend telemetry pipelines to monitor real-time token spend, rate-limit quotas, and pipeline bottlenecks; exposed operational metrics to monitoring dashboards, **reducing issue triage and debugging time from hours to minutes**.

Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

Remote

- Engineered multi-tenant backend automation microservices using **NestJS, TypeScript, and Google Workspace APIs**, orchestrating **1,000+ daily business workflows** and eliminating manual client data-entry bottlenecks.
- Built distributed workflow pipelines with **exponential backoff, idempotent webhook handlers, and rate limiting**, guaranteeing resilient job execution and preventing duplicate transaction processing across tenants.
- Developed automated verification harnesses and isolated **Docker autograder environments** for enterprise **Coursera course launches**, eliminating student code-execution regressions prior to production rollout.

Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

Bangalore, India

- Optimized high-traffic **Node.js and Express.js APIs** using **Redis response caching, PostgreSQL B-tree indexing, and connection pooling**, reducing median response time from **800ms to 320ms (60% drop)**.
- Developed candidate-job recommendation workflows utilizing multi-attribute compound indexing in **PostgreSQL**, cutting search filter latency and improving relevant role matching accuracy by **22%**.
- Built **React.js and Next.js** onboarding experiences with integrated **A/B testing**, lifting user signup conversion rates by **18%**.

Projects

WorkBridge – AI-Powered Collaboration Platform

May 2026 – Present

React.js, Next.js, Node.js, Express.js, Python, FastAPI, PostgreSQL, pgvector, Redis, OAuth | Live Project

- Architected a decoupled microservices platform featuring a **Node.js/Express API gateway** for session/auth handling and a **Python/FastAPI microservice** for vector processing.

- Built core REST APIs with **PostgreSQL and Redis**, enforcing role-based access control (RBAC) and OAuth2 authentication to achieve **150ms p95 latency** in benchmark tests.
- Implemented candidate-to-project recommendation using **Python and pgvector cosine similarity**, delivering sub-second ranking retrieval over candidate skill embeddings.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle	Aug 2026
Oracle AI Vector Search Certified Professional – Oracle	Sep 2025
Machine Learning I – Columbia University	Aug 2025
Oracle Cloud Infrastructure 2024 Generative AI Certified Professional – Oracle	Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU)	Chikkamagaluru, KA, India
Bachelor of Engineering in Mechanical Engineering	Aug 2023